# Phase 1: Controlled Intelligence Benchmark

## Controlled Environmental Interventions for Mechanistic Interpretability

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Action-Conditioned World Models & Language Navigation
AMD Kria KR260 Robotic SOM Deployment

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# 1. Controlled Capability Isolation

**Core Insight:** *Random navigation episodes confound attribution; six orthogonal capabilities isolate distinct neural mechanisms.*

## Why Random Episodes Fail

- In random episodes, high success rates may simply reflect memorized geometric shortcuts rather than genuine linguistic or spatial reasoning.
- Separating perception, relational reasoning, and collision avoidance requires **orthogonal capability benchmarks**.

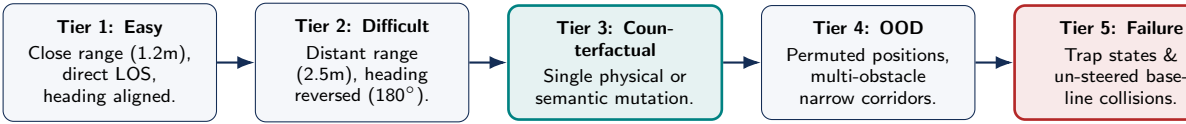
| Capability | Mechanism Under Test |
|---|---|
| **Object Grounding** | Vision-language binding |
| **Spatial Reasoning** | Relational prepositions |
| **Obstacle Avoidance** | Reactive collision circuits |
| **Action Prediction** | Forward dynamics ($s \rightarrow$ |
| **Temporal Planning** | Long-horizon waypoint s |
| **Generalization** | Zero-shot spatial invarian |

## Design Rule

Every episode tests exactly **one primary capability** while strictly logging ground-truth physical states for mechanistic analysis.

# 2. Five-Tier Systematic Difficulty Hierarchy

**Core Insight:** *Systematic variation across 5 difficulty tiers separates surface correlations from robust causal mechanisms.*




## Role of Failure Demonstrations (Tier 5)

Failure trajectories (e.g., straight-line un-steered collisions and corner traps) provide critical negative exemplars, enabling Sparse Autoencoders (Phase 4) and Causal Scrubbing (Phase 5) to isolate the exact subcircuits responsible for evasive steering.

# 3. Seven Paired Counterfactual Interventions (E1 – E7)

**Core Insight:** *Matched pair interventions* $(E_0, E_i)$ *isolate true relational invariants from superficial visual shortcuts.*

| ID | Physical Mutation | Tested Invariance |
|---|---|---|
| **E1** | Target position relocated | Preposition meaning (' |
| **E2** | Cylinder colored blue/red | Spatial vector decouple |
| **E3** | Robot heading $+90^\circ$ | Allocentric spatial orier |
| **E4** | Robot starts in quadrant II | Goal attractor invarian |
| **E5** | Obstacle inserted on LOS | Goal identity under det |
| **E6** | Cylinder replaced by box | Shape invariance |
| **E7** | Camera pitch altered $+10^\circ$ | Viewpoint robustness |

## The Disentanglement Principle

If a latent neuron fires when the target cylinder changes color but the spatial objective is invariant, that neuron encodes **appearance entanglement**.

The E1–E7 suite directly pinpoints which layers disentangle perception from action.

# 4. Resolving Fundamental Ambiguities

**Core Insight:** *Four targeted probe suites distinguish coordinate frames and separate optical appearance from kinematic reachability.*

## D1: Egocentric vs. Allocentric Left

When facing South ($\theta = -\frac{\pi}{2}$):

- **Ego Left**: East ($+x_{\text{world}}$).
- **Allocentric Left**: West ($-x_{\text{world}}$).
- Distinguishes whether the policy reasons in body or world coordinates.

## D2: Retinal Front vs. Object Behind

Moving directly forward reaches "in front", but reaching "behind" requires an active circumnavigation flank.

## D3: Apparent Size vs. Metric Proximity

A $2\times$ larger box at $2\times$ distance subtends the exact same $18.5^\circ$ FOV. Probes whether the latent space encodes metric Euclidean distance or 2D image area.

## D4: Optical Visibility vs. Reachability

Target is optically visible through a 0.15 m gap, but the AMR width (0.26 m) prohibits passage. Probes topological planning.

# 5. Partitioning Strategy: Zero Observation Leakage

**Core Insight:** *Splitting strictly by environmental configuration seeds guarantees zero observation or frame leakage across splits.*

## The Danger of Frame-Level Random Splits

Randomly shuffling transitions $(o_t, a_t, o_{t+1})$ leaks adjacent video frames across train and test sets, artificially inflating generalization metrics.

## Partitioning by Environment Configuration

Splits are strictly segregated by **entire scene configurations and random seeds**:

- **Train Benchmark**: 408 episodes (12,595 transitions).
- **Validation Benchmark**: 96 episodes (2,911 transitions).
- **Held-out Test Benchmark**: 96 episodes (2,885 transitions).

**Total Benchmark Corpus**

| Dataset Split | Episodes | Frames |
|---|---|---|
| Train Benchmark | 408 | 12,595 |
| Val Benchmark | 96 | 2,911 |
| Held-out Test | 96 | 2,885 |
| Counterfactual (E1-E7) | 350 | 12,250 |
| Disambiguation | 100 | 3,254 |
| **Total Corpus** | **1,050** | **33,895** |

Total Storage: 165.78 MB (Compressed NPZ)

# 6. Phase 1 Verification & Prerequisites for Phase 2

**Core Insight:** *With 1,050 reproducible episodes across 6 capabilities and 5 tiers, Phase 1 acceptance criteria are fully satisfied.*

## Phase 1 Acceptance Criteria Checklist

- ✓ **Reproducible Episodes**: 1,050 episodes generated deterministically ($> 100$ per capability).
- ✓ **Controlled Variations**: Controlled variations across position, orientation, color, shape, and obstacles.
- ✓ **Split Separation**: Train, validation, held-out test, counterfactual, and disambiguation splits completely isolated.
- ✓ **Ground-Truth Physical State**: 15-field schema logs exact poses, velocities, and 3D coordinates for mechanistic probing without leakage to models.
- ✓ **Data Integrity**: 0 NaNs, verified tensor dimensions, and complete companion metadata.

## Readiness for Phase 2

The experimental laboratory is fully prepared to execute **Phase 2: Establish Behavioral Baselines** across Baseline Policy, LeWM, and H-JEPA under identical evaluation seeds.